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GRADO EN FÍSICA

Práctica de Complejos

Quantum Chaos and the Statistics of Energy Levels

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Abstract

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Índice

1 Introduction

The study of complex systems allows to explore the transition between the classical order and the statistichal manifestations in the quantum regime. As in classical mechanics chaos isdefined as the exponential sensitivity to initial conditions, the linearity in the Schrödinger equation does not allow for a direct translation of this definition to quantum mechanics. However, the phenomena known as 'Quantum Chaos' is manifested throughout the statistichal properties of the energy spectrum of the system.

The nuclei in this project is the Bohigas-Giannoni-Schmit (BGS) conjecture, which states that the energy levels of a quantum system whose classical counterpart is chaotic will exhibit the same statistical properties as the eigenvalues of a random matrix ensemble.

To validate this hipothesis, we'll investigate the distribution of spacing between adjacent energy levels in two systems (s_i): one integrable and one chaotic.

The first one is the two-dimensional **rectangular billiard**, which is an **integrable system**. In this system, the random matrix theory (RMT) predicts that the distribution of level spacings follows a Poisson distribution, which is given by:

$$P(s) = e^{-s} \tag{1.1}$$

The second system is the **stadium billiard**, which is a **chaotic system**. In this case, RMT predicts that the distribution of level spacings follows the Wigner-Dyson distribution, which is given by:

$$P(s) = \frac{\pi}{2} s e^{-\frac{\pi}{4}s^2} \tag{1.2}$$

where s is the normalized spacing between adjacent energy levels in both cases.

To integrate the time independent Schrödinger equation for these systems, we will use the **Finite Difference Method** (FDM): this method allows us to discretize the spatial domain and solve for the energy eigenvalues and eigenfunctions of the system.

2 Methods

To be able to solve the Schrödinger's Equation $H\psi = E\psi$, we'll first need to discretize the space and then solve the resulting eigenvalue problem. We'll need to impose the Dirichlet conditions to the domain, which means that the wavefunction ψ must vanish at the boundaries of the billiard.

$$\psi|_{\partial\Omega} = 0 \tag{2.1}$$

All the methods used in this project are implemented in Python, and the code is available in the GitHub repository . To explain the followed procedure, we'll first introduce the Mascara method, which is used to discretize the system, and then the Finite Difference Method, which is used to solve the resulting eigenvalue problem.

2.1 Discretisation of the system: Mascara Method

To easily create different geometries, we decided to implement a finite difference method that takes different masks and computes the corresponding eigenvalues. The mask represents a grid containing the geometry of the problem. For each pixel, we determine whether it belongs to the geometry by assigning it a true or false value.

To carry out this project, we created three different masks: a rectangular mask, a stadium-shaped mask, and a quarter-stadium mask. The following figure shows these masks:

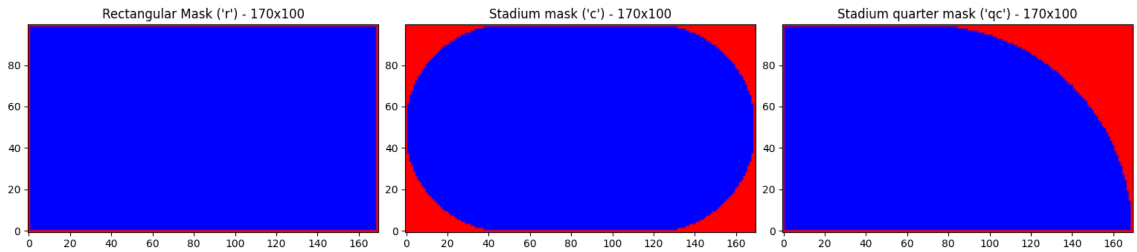


Figura 1: Different mask geometries used in this experiment. From left to right: a rectangular mask, a stadium-shaped mask, and a quarter-stadium mask.

The project aimed to investigate the stadium geometry, which produces chaotic behavior in the eigenvalue spectrum. However, we encountered a problem: the stadium has two symmetries, one with respect to the x-axis and another with respect to the y-axis. In Chapter 1.4 of the thesis by *Wisniacki, Diego A.*, it is explained that the eigenvalues are divided into four different sets because of these symmetries:

- even | even
- even | odd
- odd | even
- odd | odd

This produces four different Wigner–Dyson distributions, which, when combined, result in a Poisson distribution. By removing the symmetries and considering only a quarter of the stadium, the chaotic behavior becomes visible. [1]

2.2 Finite Difference Method

Once we have a grid of the system, where $x_i = i\Delta x$ and $y_j = j\Delta y$, being $i = 1, 2, ...n_x$ and $j = 1, 2, ...n_y$. As we know, the HAmiltonian for a teo-dimensional system is given by:

$$H = -\frac{\hbar^2}{2m}\left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2}\right) + V(x, y) \tag{2.2}$$

Since in our systems we have a free particle, the potential $V(x,y)$ is zero everywhere inside the billiard and infinite outside. With this consideration, the Schrödinger equation can be rewritten as the Helmholtz equation:

$$\left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + k^2\right)\psi(x,y) = 0 \tag{2.3}$$

where $k^2 = E$ (using units such that $\hbar^2/2m = 1$).
Now, we can discretize the second derivatives using finite differences. If we consider one of the dimensions, for example the x-direction, we can approximate the wave function in the nearest points with a Taylor expansion:

$$\psi(x_i \pm \Delta x) = \psi(x_i) \pm \Delta x \psi'(x_i) + \frac{\Delta x^2}{2}\psi''(x_i) \pm O(\Delta x^3) \tag{2.4}$$

Same in the y-direction. If we sum the two equations and subtract them, we can get an expression for the second derivative:

$$\psi''(x_i) = \frac{\psi(x_i + \Delta x) - 2\psi(x_i) + \psi(x_i - \Delta x)}{\Delta x^2} + O(\Delta x^2) \approx \frac{\psi_{i+1} - 2\psi_i + \psi_{i-1}}{\Delta x^2} \tag{2.5}$$

Taking this in consideration, and applying the same procedure for the y-direction, we'll apply the Helmholtz equation in its discretized form 2.6 to the point (x_i, y_j) of the grid.

$$\left(\frac{\psi_{i+1,j} - 2\psi_{i,j} + \psi_{i-1,j}}{\Delta x^2} + \frac{\psi_{i,j+1} - 2\psi_{i,j} + \psi_{i,j-1}}{\Delta y^2} + k^2\psi_{i,j}\right) = 0 \tag{2.6}$$

Is more efficient to write the resulting equation in a matrix form, where the matrix H is given by:

$$H' = \begin{pmatrix} D & \frac{1}{\Delta x^2} & 0 & \dots & \frac{1}{\Delta y^2} & 0 & \dots \\ \frac{1}{\Delta x^2} & D & \frac{1}{\Delta x^2} & 0 & \dots & \frac{1}{\Delta y^2} & \dots \\ 0 & \frac{1}{\Delta x^2} & D & \frac{1}{\Delta x^2} & 0 & \dots & \dots \\ \vdots & 0 & \ddots & \ddots & \ddots & 0 & \vdots \\ \frac{1}{\Delta y^2} & \vdots & 0 & \frac{1}{\Delta x^2} & D & \frac{1}{\Delta x^2} & 0 \\ 0 & \frac{1}{\Delta y^2} & \vdots & \dots & \frac{1}{\Delta x^2} & D & \frac{1}{\Delta x^2} \\ \vdots & \dots & \dots & \dots & 0 & \frac{1}{\Delta x^2} & D \end{pmatrix} \tag{2.7}$$

This matrix is called the finite difference matrix (FDM) and it is a sparse matrix, since most of its elements are zero. It has a total of $n_x \times n_y$ elements. The diagonal elements are given by $D = k^2 - \left(\frac{2}{\Delta x^2} + \frac{2}{\Delta y^2}\right)$. The second diagonal elements are going by $\frac{1}{\Delta x^2}$ and it corresponds to the coupling between the nearest points in the x-direction. The elements that are n_x positions away from the diagonal are given by $\frac{1}{\Delta y^2}$ and it corresponds to the coupling between the nearest points in the y-direction. Since in our systems we have a free particle, the potential $V(x, y)$ is zero everywhere inside the billiard and infinite outside.

3 Results

3.1 Statistichal analysis of spacing distribution

First,we calculate the first 10000 eigenvalues of each system by diagonalizing the sparse matrix that we obtained applying the DFM 2.7. Then, with those arrays, we compute a new one that has the spacing distribution between consecutive eigenvalues, this is $s = E_{i+1} - E_i$. Finally, we plot an histogram with those, as it follows:

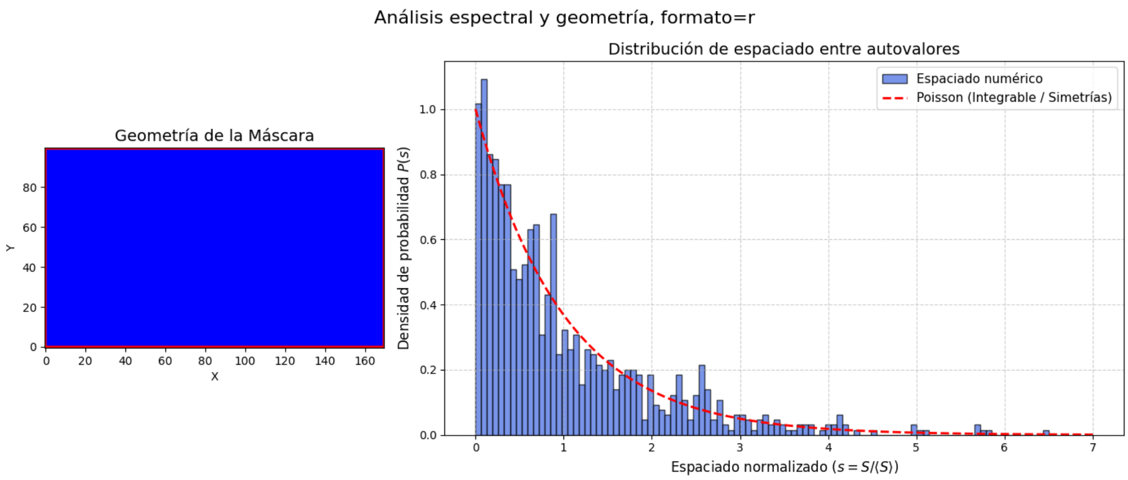


Figura 2: Poisson distribution for the integrable rectangle.

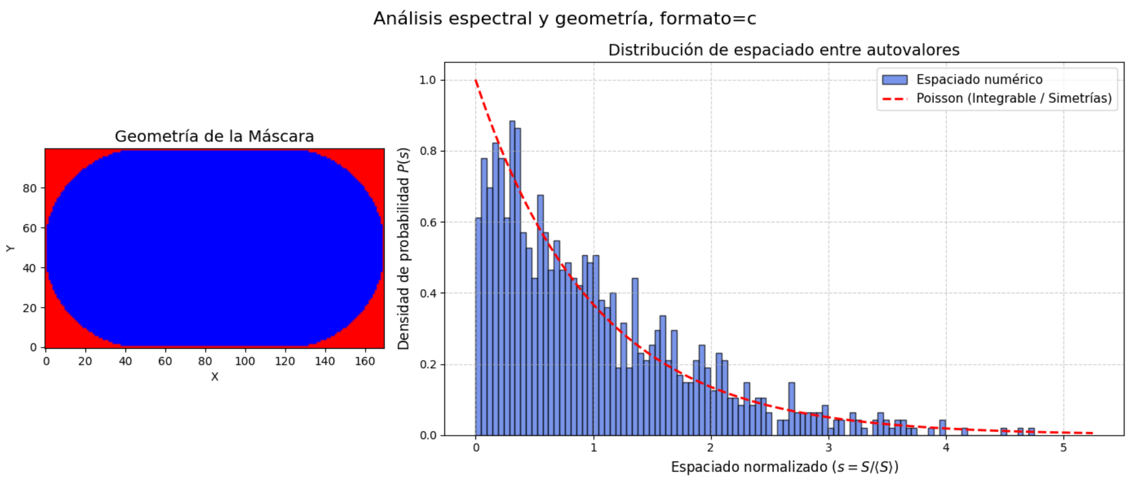


Figura 3: Poisson distribution for the non-integrable complete stadium.

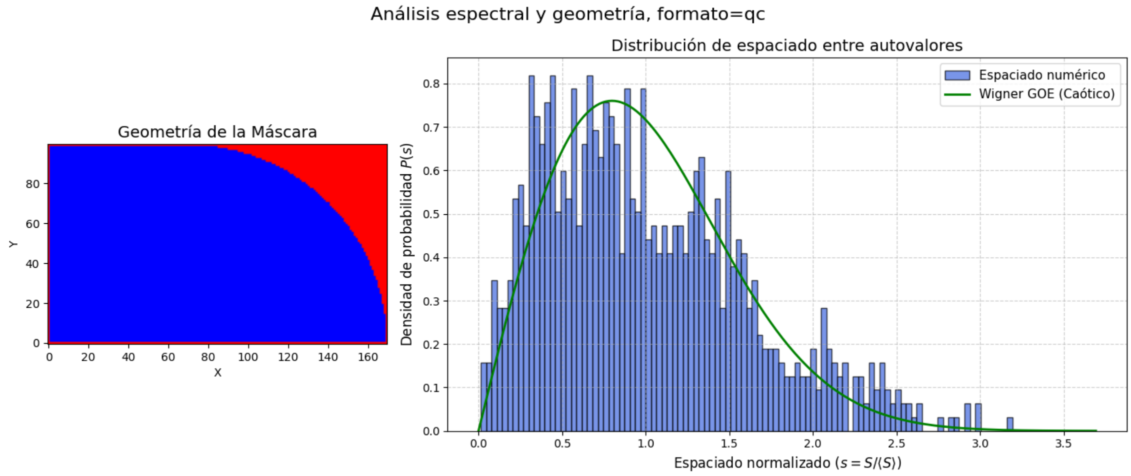


Figura 4: Wigner-Dyson distribution for a quarter of the non-integrable stadium.

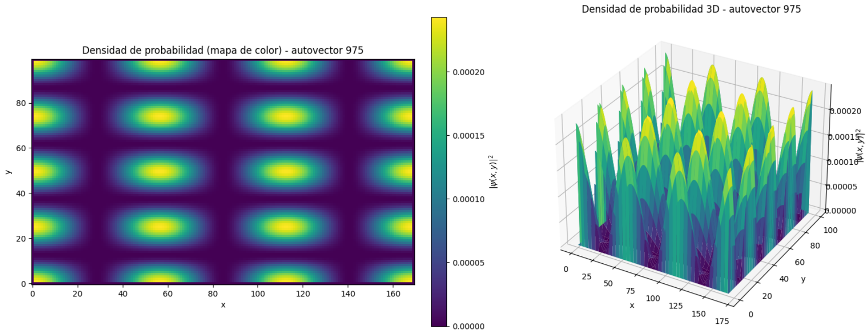
As expected, we can clearly see in Figure 2 that the spacing distribution for the integrable rectangle follows a Poisson distribution. This graphic shows how this systems, even if it is integrable, that their eigenvalues are not correlated, then the probability of finding two eigenvalues close to each other is high ($P(s) \to 1$ when $s \to 0$). Also, in the case of the non-integrable billiad (the complete stadium showed in Figure 3), we can see that it also follows a near Poisson distribution, due to the symmetries of the system. This symmetries make the systm to have four pairs of solutions, then, the degenerate levels will increase the probability of finding two eigenvalues close to each other, even if the system is non-integrable.

On the other hand, if we eliminate the $x - axis$ and $y - axis$ simmetries, having just a quarter of the stadium as we see in Figure 4, it is noticeable that the spacing distribution for the non-integrable billiard follows a Wigner-Dyson distribution. In this case, the opposite happens: the system may seem chaotic, but its eigenvalues are correlated, in a way that makes them repel each other or, in other words, the probability of finding two near-degenerate eigenvalues is very low ($P(s) \to 0$ when $s \to 0$).

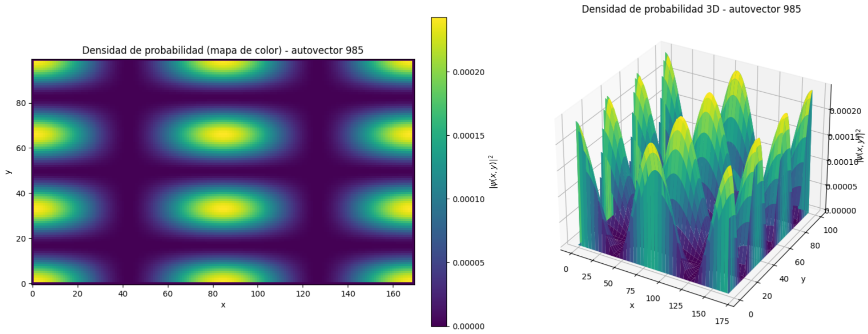
3.2 High-energy states visualization

To determine the chaocity of the system, we must look at high-energy states, and compare them to the non-chaotic system. This is due to the fact that, in the low-energy states, the eigenvectors may seem like those of a string, therrefore, they don't contain the necessary information to see if the system is chaotic or not. We then plot the probability density of some of the high-energy states for both systems.

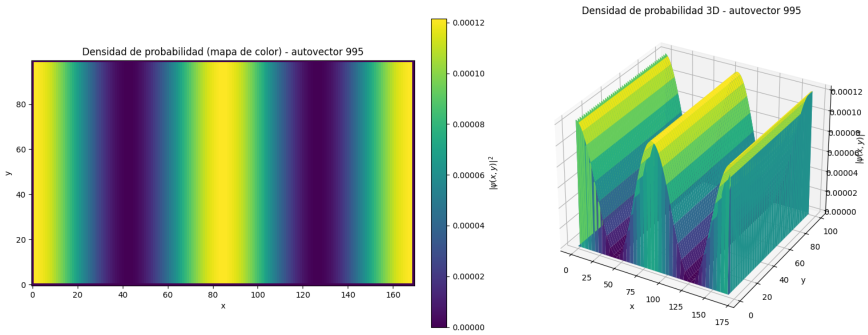
The three graphics of Figure 5 show the probability density for three of the high-energy states of the integrable rectangle.We see that the patterns are very regular, and they look like the fundaental modes of a string, with nodes and antinodes. We can determine then that this system is not chaotic, as expected, since there are



(a) Probability density for the eigenvalue 975.



(b) Probability density for the eigenvalue 985.

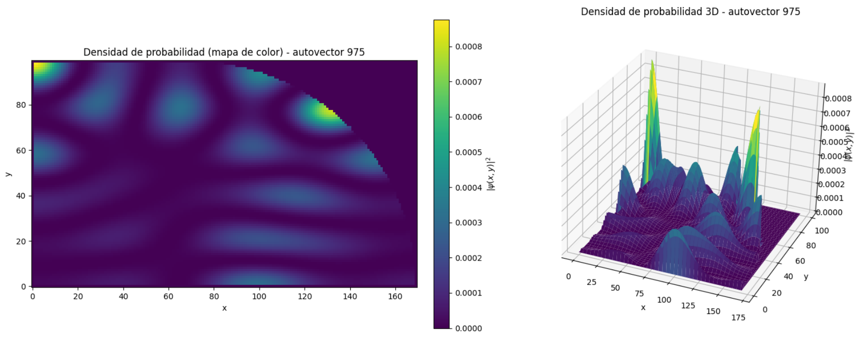


(c) Probability density for the eigenvalue 995.

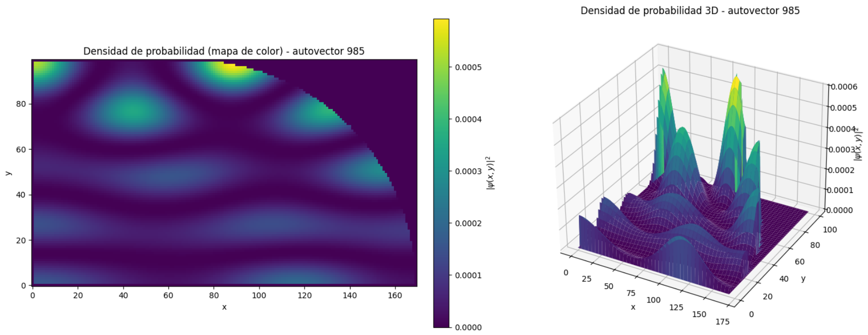
Figura 5: Visualization of various of the high-energy states for the **integrable rectangle**. The probability density is represented in a color scale, where the red areas correspond to higher probability density and the blue areas correspond to lower probability density.

periodic patterns in the probability density.

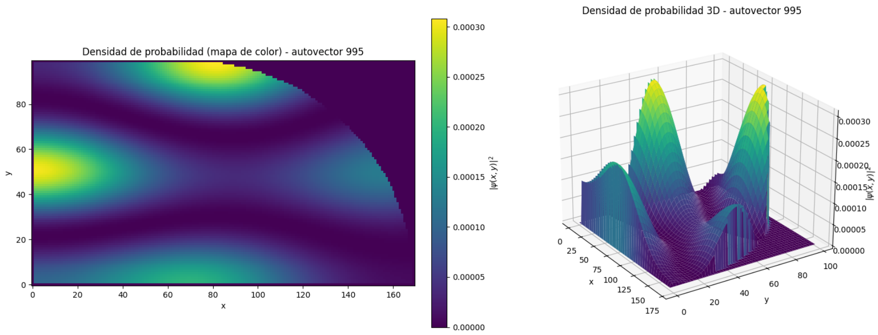
In contrast, the same eigenvalues are shown in Figure 6. Is important to note that we only represented a quarter of stadium, because is the significant case in our study. In this system, the probability density shows 'quantum scars' or trajectories, wich corresponds to classical periodic orbits that a particle would follow within the stadium. Instead of a uniform distribution, these scars represent a quantum localization along specific geometric paths.



(a) Probability density for the eigenvalue 975.



(b) Probability density for the eigenvalue 985.



(c) Probability density for the eigenvalue 995.

Figura 6: Visualization of various of the high-energy states for the **non-integrable quarter billiard**. The probability density is represented in a color scale, where the red areas correspond to higher probability density and the blue areas correspond to lower probability density.

4 Conclusions

Referencias

[1] Diego Ariel Wisniacki. *Caos cuántico en sistemas dependientes de un parámetro*. PhD thesis, Universidad de Buenos Aires. Facultad de Ciencias Exactas y Naturales, 1999.